Predicting Vaccination Rates Using Behavioral Data

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Project Goal and Motivation

- Goal: predict vaccination probabilities for two outcomes using survey responses:
  - H1N1 vaccine uptake
  - Seasonal flu vaccine uptake
- Data source: National 2009 H1N1 Flu Survey (NHFS), collected during the 2009–2010 pandemic season.
- Public health motivation: identify characteristics associated with likely and unlikely vaccine uptake to inform outreach planning.

Challenge Framing and Evaluation

- This is a multilabel prediction task with two binary labels:
  - `h1n1_vaccine` and `seasonal_vaccine`
- Outputs must be probabilities in $[0, 1]$, not hard class labels.
- Planned model comparison metric: macro ROC AUC (mean AUC across both labels).
- Proposal scope: model design, evaluation plan, and expected interpretation (no final empirical claims yet).

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Dataset Structure and Feature Groups

- Sample size: about 26,707 respondents, with 35 predictors plus respondent identifier.
- Predictor groups:
  - Behavioral actions (e.g., mask use, handwashing, avoidance)
  - Attitudes and risk beliefs (e.g., concern, perceived vaccine effectiveness)
  - Access and context (e.g., doctor recommendation, insurance)
  - Demographics and geography (age, education, region, employment)
- Mixed data types: binary, ordinal, nominal categorical, and count variables.

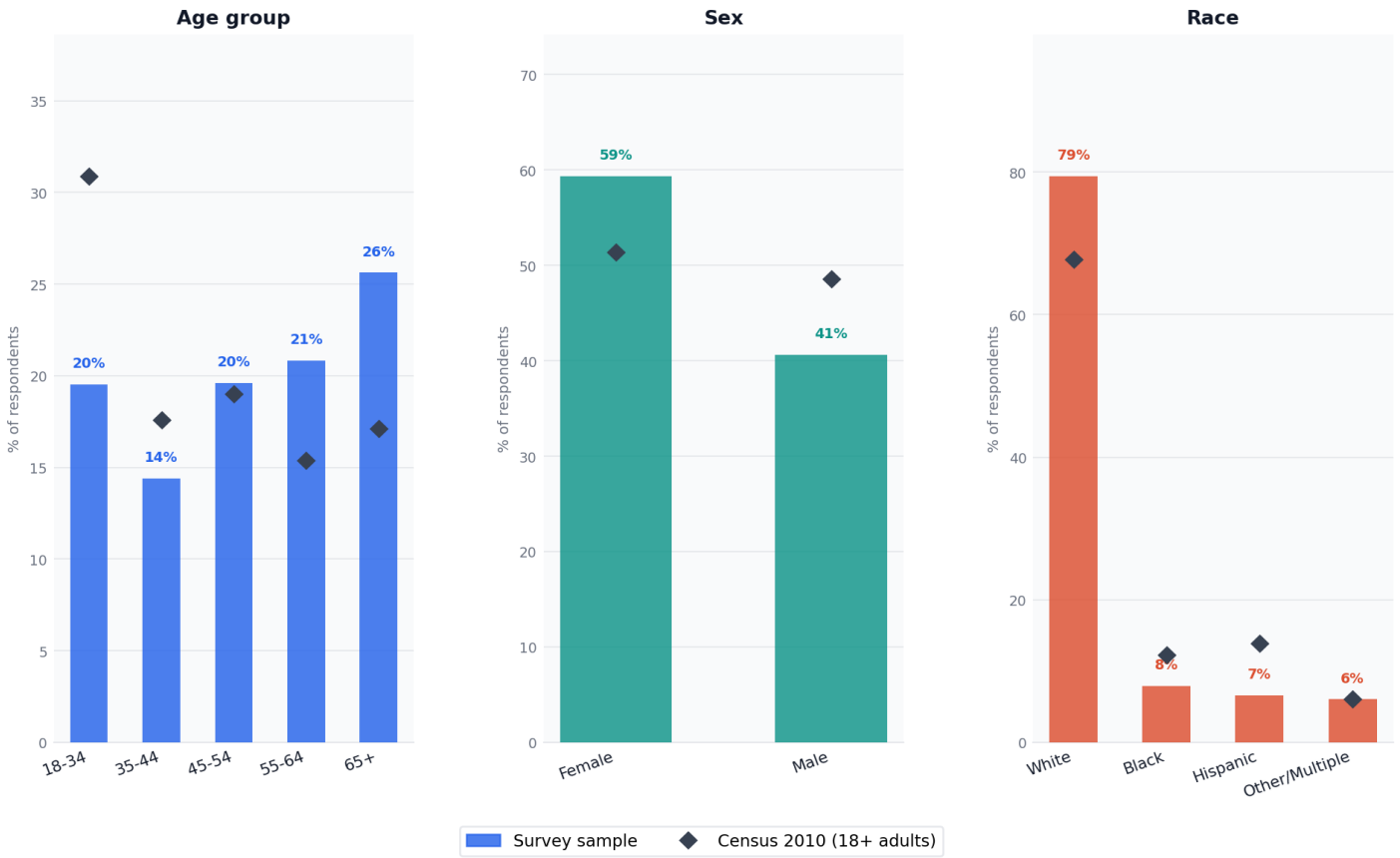
Planned Preprocessing and Validation Strategy

- Missing data plan:
  - Use explicit missing-category treatment for key categorical fields.
  - Use structured imputation for selected predictors where justified.
- Encoding plan:
  - Keep ordinal scales ordered where meaningful.
  - Use indicator encoding for nominal categories.
- Evaluation split plan:
  - Stratified cross-validation on both outcomes for robust model comparison.
  - Report macro ROC AUC and probability calibration diagnostics.

H1N1 Exploratory Data Analysis

Figure 1 — Sample Composition: Survey vs. U.S. Census Baseline

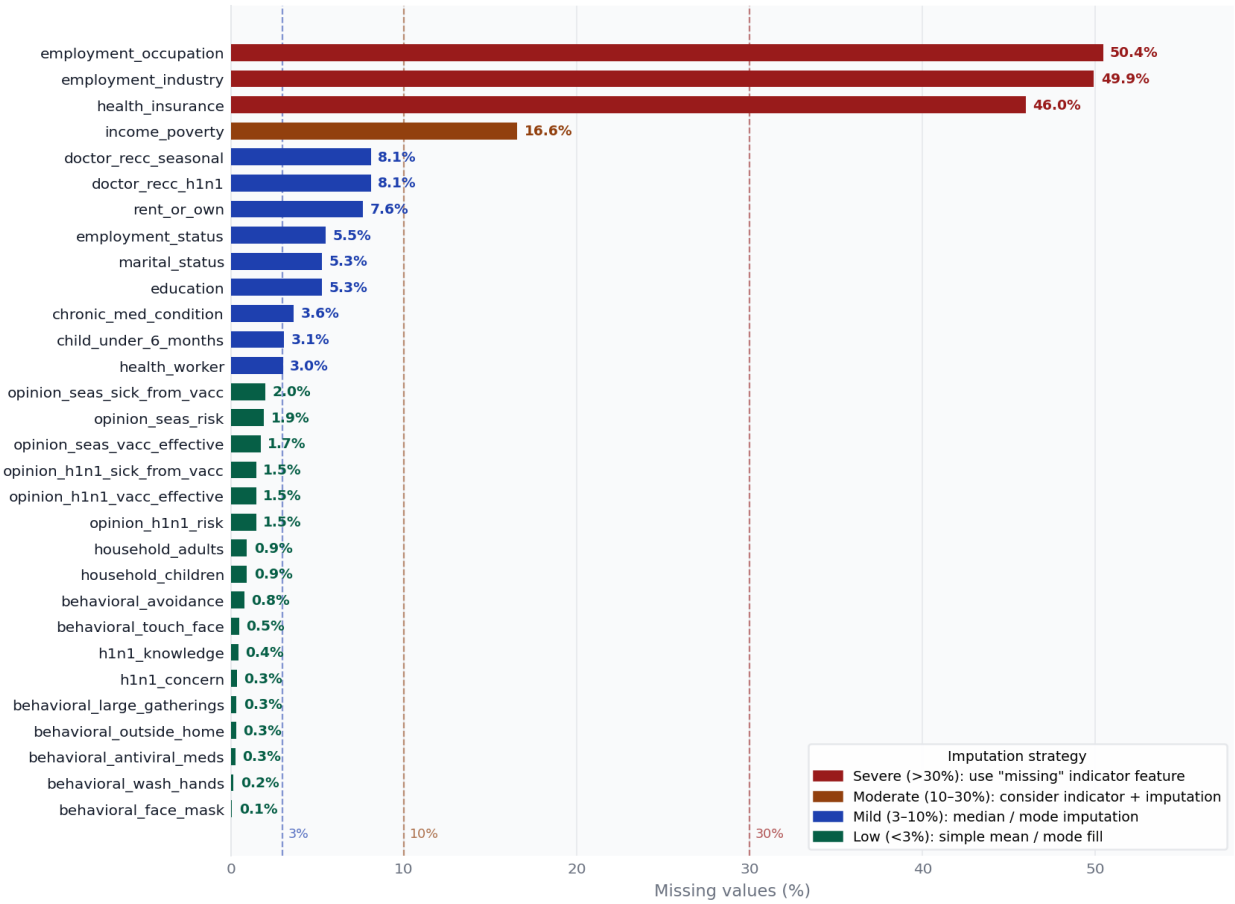
Each panel compares the share of respondents in the survey sample (filled bar) against the U.S. Census 2010 baseline (◆ marker) for the 18+ adult population.
Source: U.S. Census Bureau, "2008 National Population Projections," Tables 10 & 11.



H1N1 Feature Relationships

Figure 2 — Missing Data Profile (Training Set, n = 26,707)

Features ranked by missingness percentage. Colour tier indicates imputation strategy: severe → categorical "missing" indicator; moderate/mild → median/mode imputation; low → simple mean/mode fill.



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Joint Multivariate Hierarchical Model

For respondent i in group $g(i)$, and outcome $k \in \{\text{H1N1}, \text{Seasonal}\}$:

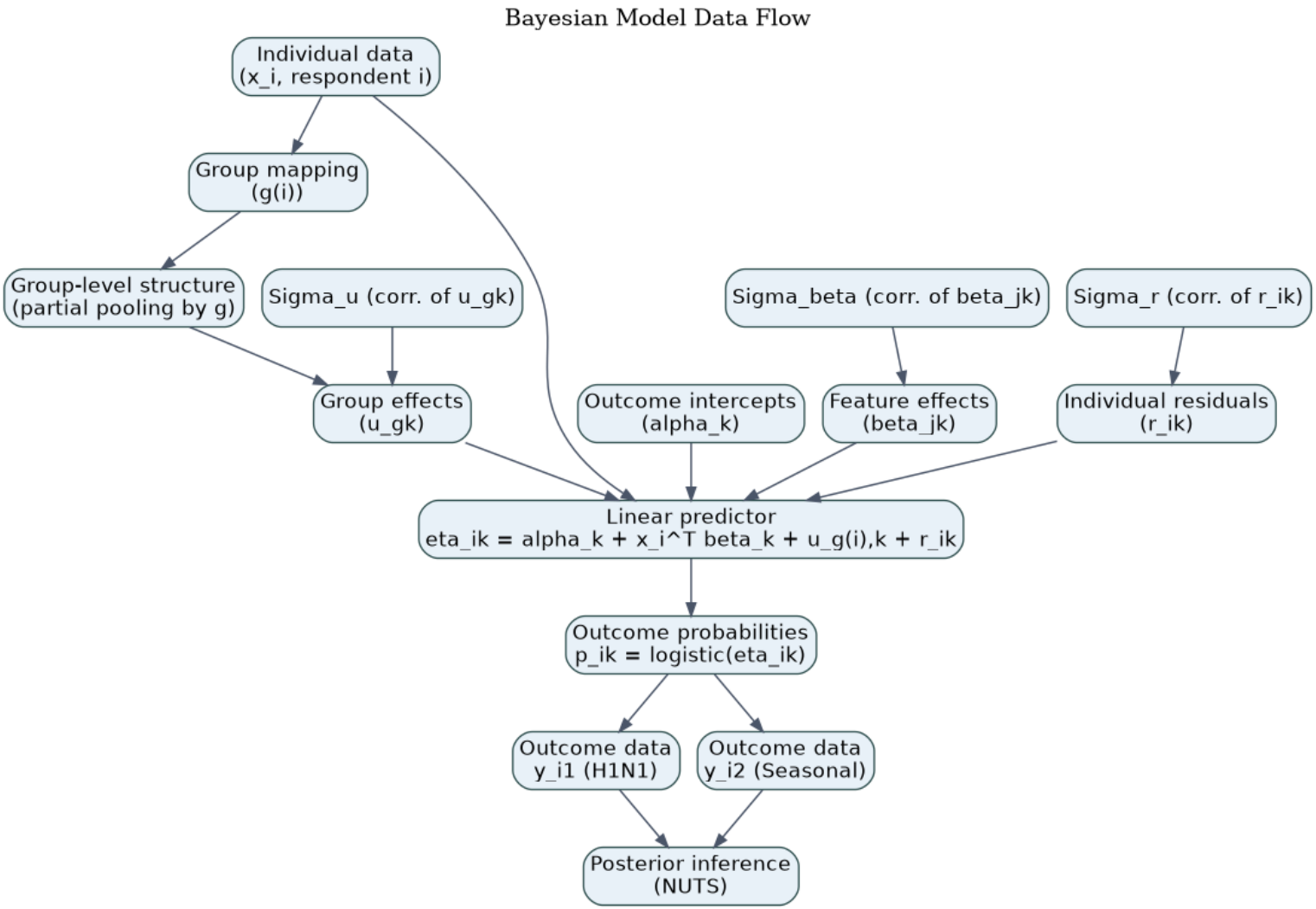
$$\text{logit}(p_{ik}) = \alpha_k + \mathbf{x}_i^\top \boldsymbol{\beta}_k + u_{g(i),k} + r_{i,k}, \qquad y_{ik} \sim \text{Bernoulli}(p_{ik})$$

$$\mathbf{u}_g = (u_{gk}) \sim \mathcal{N}(\mathbf{0}, \Sigma_u), \quad \mathbf{r}_i = (r_{ik}) \sim \mathcal{N}(\mathbf{0}, \Sigma_r)$$

$$(\beta_{jk}) \sim \mathcal{N}(\mathbf{0}, \Sigma_\beta), \qquad \Sigma = DRD, \; R \sim \text{LKJ}(\eta)$$

- Directly models cross-outcome dependence between the two vaccine outcomes.
- Partial pooling allows stable estimates for smaller subgroups.
- Candidate group structure: HHS region and selected demographic strata.

Data and Model Flow



Inference and Model-Checking Plan

- Inference approach: MCMC posterior sampling (NUTS) for the joint model.
- Planned diagnostics:
  - Convergence checks with $\hat{R}$ near 1.00 and sufficient effective sample size.
  - Posterior predictive checks for calibration and distributional fit.
  - Sensitivity analysis under alternative prior and LKJ correlation scales.
- Outputs to interpret:
  - Posterior means/intervals for key predictors.
  - Group-level random effect uncertainty and cross-outcome correlations.
  - Predicted vaccination probabilities with credible intervals.

Planned Baselines and Comparison Criteria

- Baseline 1: regularized logistic regression.
- Baseline 2: random forest classifier.
- Proposed comparison dimensions:
  - Predictive performance: macro ROC AUC across both vaccine targets.
  - Calibration quality of predicted probabilities.
  - Interpretability and uncertainty quantification for public health use.
- Hypothesis: joint multivariate Bayesian model will offer stronger uncertainty-aware interpretation with competitive predictive performance.

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Expected Findings and Interpretation Plan

- We expect stronger vaccination probability when respondents report:
  - doctor recommendation,
  - higher perceived risk,
  - higher perceived vaccine effectiveness,
  - healthcare worker status.
- We expect behavioral prevention variables to provide additional predictive signal.
- Interpretation emphasis:
  - quantify directional effects and uncertainty using credible intervals,
  - compare consistency of patterns across H1N1 and seasonal outcomes,
  - identify subgroups where uncertainty remains large.

Limitations, Ethics, and Next Steps

- Known data limitations:
  - self-report bias and recall error,
  - non-response bias from phone-survey collection,
  - no direct medical verification in this dataset.
- Ethical interpretation plan:
  - avoid causal claims from observational survey data,
  - examine subgroup uncertainty before policy recommendations.
- Next project milestones: finalize preprocessing, fit planned models, complete posterior checks, and prepare validated comparison results.